

Rio Yokota

CONTACT INFORMATION

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RESEARCH INTERESTS

High performance computing; Large Scale Deep Learning; Numerical Analysis

EDUCATION

Keio University, Yokohama, Japan

Ph.D., Mechanical Engineering, July, 2009

- Dissertation Topic: “Validation of Vortex Methods as a Direct Numerical Simulation of Turbulence”
- Advisor: Shinnosuke Obi

HONORS AND AWARDS

ACM Gordon Bell prize, price/performance (2009)

ACADEMIC EXPERIENCE

- | | |
|-------------------------|--|
| 10/2024 - | Professor, Institute of Science Tokyo |
| 1/2023 - 9/2024 | Professor, Tokyo Institute of Technology |
| 4/2015 - 12/2022 | Associate Professor, Tokyo Institute of Technology |
| 9/2011 - 3/2015 | Research Scientist, KAUST |
| 9/2010 - 8/2011 | Post-doctoral Researcher, Boston University |
| 2/2009 - 8/2010 | Post-doctoral Researcher, University of Bristol |
| 4/2007 - 2/2009 | Research Assistant, Keio University |

BOOK CHAPTERS

1. R. Yokota, M. AbdulJabbar, *N*-body methods on Xeon Phi coprocessors, *High Performance Parallelism Pearls* (2014)
2. R. Yokota and L. A. Barba, Treecode and fast multipole method for N-body simulation with CUDA, *GPU Computing Gems* (2011)

PEER REVIEWED JOURNALS

1. Satoshi Yui, Hiromichi Kobayashi, Makoto Tsubota, Rio Yokota, Quantum turbulence coupled with externally driven normal-fluid turbulence in counterflow of superfluid ^4He , *Journal of the Physical Society of Japan*, 94, 043601 (2025).
2. Takuya Akiba, Makoto Shing, Yujin Tang, Qi Sun, David Ha, Evolutionary Optimization of Model Merging Recipes, *Nature Machine Intelligence* (2025).
3. Kenta Niwa, Hiro Ishii, Hiroshi Sawada, Akinori Fujino, Noboru Harada, Rio Yokota, ,Natural Gradient Primal-Dual Method for Decentralized Learning, *Transactions on Signal and Information Processing over Networks*, (2024).
4. Qianxiang Ma, Rio Yokota, An Inherently Parallel $\mathcal{H}^2$ -ULV Factorization for Solving Dense Linear Systems on GPUs, *International Journal of High Performance Computing Applications*, (2024).

5. Hiroyuki Ootomo, Katsuhisa Ozaki, Rio Yokota, DGEMM on Integer Matrix Multiplication Unit, *International Journal of High Performance Computing Applications*, (2024).
6. Erik Daxberger, Siddharth Swaroop, Kazuki Osawa, Rio Yokota, Richard E. Turner, Jose Miguel Hernandez-Lobato, Mohammad Emtiyaz Khan, Improving Continual Learning by Accurate Gradient Reconstructions of the Past, *Transactions on Machine Learning Research*, (2023).
7. Hiroki Naganuma, Kartik Ahuja, Shiro Takagi, Tetsuya Motokawa, Rio Yokota, Kohta Ishikawa, Ikuro Sato, Ioannis Mitliagkas, Empirical Study on Optimizer Selection for Out-of-Distribution Generalization, *Transactions on Machine Learning Research*, (2023).
8. Sameer Deshmukh, Rio Yokota, George Bosilca, Cache Optimization and Performance Modeling of Batched, Small, and Rectangular Matrix Multiplication on Intel, AMD, and Fujitsu Processors, *ACM Transactions on Mathematical Software*, (2023).
9. Muhammad Ridwan Apriansyah, Rio Yokota, Parallel QR Factorization of Block Low-Rank Matrices, *ACM Transactions on Mathematical Software*, (2022). <https://doi.org/10.1145/3538647>
10. Hiroyuki Ootomo, Rio Yokota, Recovering Single Precision Accuracy from Tensor Cores While Surpassing the FP32 Theoretical Peak Performance, *The International Journal of High Performance Computing Application*, (2022).
11. Tingyu Wang, Rio Yokota, Lorena A. Barba, ExaFMM: a high-performance fast multipole method library with C++ and Python interfaces, *The Journal of Open Source Software*, Vol. 6, No. 61, pp. 3145 (2021).
12. Kazuki Osawa, Yohei Tsuji, Yuichiro Ueno, Akira Naruse, Chuan-Sheng Foo, Rio Yokota, Scalable and Practical Natural Gradient for Large-Scale Deep Learning, *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2020); PP:10.1109/TPAMI.2020.3004354.
13. Davoud S. Shamshirgar, Rio Yokota, Anna-Karin Tornberg, and Berk Hess, Regularizing the Fast Multipole Method for use in Molecular Simulation, *The Journal of Chemical Physics*, 151, 234113 (2019)
14. Akihiro Ida, Hiroshi Nakashima, Tasuku Hiraishi, Ichitaro Yamazaki, Rio Yokota, Takeshi Iwashita, QR Factorization of Block Low-rank Matrices with Weak Admissibility Condition, *Journal of Information Processing*, Vol. 12, No. 4 (2019)
15. Ichitaro Yamazaki, Akihiro Ida, Rio Yokota, Jack Dongarra, Distributed Memory Lattice H-matrix Factorization, *The International Journal of High Performance Computing Applications* (2019).
16. Mustafa Abduljabbar, Mohammed Al Farhan, Noha Al-Harthi, Rui Chen, Rio Yokota, Hakan Bagci, David Keyes, Extreme Scale FMM-Accelerated Boundary Integral Equation Solver for Wave Scattering, *SIAM Journal on Scientific Computing*, Vol. 41, No. 3, pp. C245–C268 (2019).
17. Naoya Maruyama, Takayuki Aoki, Kenjiro Taura, Rio Yokota, Mohamed Wahib, Motohiko Matsuda, Keisuke Fukuda, Takashi Shimokawabe, Naoyuki Onodera, Michel Müller, Shintaro Iwasaki, Highly Productive, High-Performance Application Frameworks for Post-Petascale Computing, *Advanced Software Technologies for Post-Peta Scale Computing*, pp. 77–98 (2018)
18. Rio Yokota, Huda Ibeid, David Keyes, Fast Multipole Method as a Matrix-Free Hierarchical Low-Rank Approximation *Lecture Notes in Computational Science and Engineering*, Springer, Vol. 117, pp. 227-244 (2017)
19. Huda Ibeid, Rio Yokota, Jennifer Pestana, David Keyes, Fast Multipole Preconditioners for Sparse Matrices Arising from Elliptic Equations, *Computing and Visualization in Science* (2017). <https://doi.org/10.1007/s00791-017-0287-5>
20. Rio Yokota, Communication Optimization of Distributed Memory FMM for Large Scale Boundary Element Methods, *Simulation*, Vol. 35, No. 3, pp. 23–29 (2016)

21. Rio Yokota, Tradeoff between FMM and $\mathcal{H}^2(HSS)$ -matrices, *Journal of the Japan Society for Computational Engineering and Science*, Vol. 21, No. 4, pp. 3498–3501 (2016)
22. Julio E. Castrillon-Candas, Marc G. Genton, Rio Yokota, Multi-level Restricted Maximum Likelihood Covariance Estimation and Kriging for Large Non-gridded Spatial Datasets, *Spatial Statistics*, Vol. 18, pp. 105–124 (2016)
23. Huda Ibeid, Rio Yokota, David Keyes, A Performance Model for the Communication in Fast Multipole Methods on HPC Platforms, *International Journal of High Performance Computing Applications*, Vol. 30, No. 4, pp. 423–437 (2016)
24. Abdelhalim Amer, Satoshi Matsuoka, Miquel Pericàs, Naoya Maruyama, Kenjiro Taura, Rio Yokota, Pavan Balaji, Scaling FMM with Data-Driven OpenMP Tasks on Multicore Architectures, *Lecture Notes in Computer Science*, Vol. 9903 LNCS, pp. 156–170 (2016).
25. Rio Yokota, George Turkiyyah, David Keyes, Communication Complexity of the Fast Multipole Method and its Algebraic Variants, *Supercomputing Frontiers and Innovations*, Vol. 1, No. 1, pp. 63–84 (2014)
26. Yousuke Ohno, Rio Yokota, Hiroshi Koyama, Gentaro Morimoto, Aki Hasegawa, Gen Masumoto, Noriaki Okimoto, Yoshinori Hirano, Huda Ibeid, Tetsu Narumi, Makoto Taiji, Petascale Molecular Dynamics Simulation Using the Fast Multipole Method on K Computer, *Computer Physics Communications*, Vol. 185, No. 10, pp. 2575–2585 (2014)
27. Hatem Ltaief, Rio Yokota, Data-Driven Execution of Fast Multipole Methods, *Concurrency and Computation: Practice and Experience*, Vol. 26, No. 11, pp. 1935–1946 (2014)
28. Rio Yokota, An FMM Based on Dual Tree Traversal for Many-Core Architectures, *Journal of Algorithms and Computational Technology*, Vol. 7, No. 3, pp. 301–324 (2013)
29. Rio Yokota, Lorena A. Barba, Tetsu Narumi, Kenji Yasuoka, Petascale Turbulence Simulation Using a Highly Parallel Fast Multipole Method, *Computer Physics Communications*, Vol. 184, No. 3, pp. 445–455 (2013)
30. Rio Yokota, Lorena A. Barba, FMM-based Vortex Method for Simulation of Isotropic Turbulence on GPUs, Compared with a Spectral Method, *Computers and Fluids*, Vol. 80, pp. 17–27 (2013)
31. Rio Yokota, Lorena A. Barba, A Tuned and Scalable Fast Multipole Method as a Preeminent Algorithm for Exascale Systems, *International Journal of High Performance Computing Applications*, Vol. 26, No. 4, pp. 337–346 (2012)
32. Rio Yokota, FMM Tree Construction on GPUs, *Ensemble*, Vol. 14, No. 2, pp. 85–89 (2012)
33. Rio Yokota, Lorena A. Barba, Hierarchical N-body Simulations with Auto-tuning for Heterogeneous Systems, *Computing in Science and Engineering*, Vol. 14, No. 3, pp. 30–39 (2012)
34. Rio Yokota, Jaydeep P. Bardhan, Matthew G. Knepley, Lorena A. Barba, Tsuyoshi Hamada, Biomolecular Electrostatics Using a Fast Multipole BEM on up to 512 GPUs and a Billion Unknowns, *Computer Physics Communications*, Vol. 182, No. 6, pp. 1272–1283 (2011)
35. Rio Yokota and Lorena A. Barba, Comparing the Treecode with FMM on GPUs for Vortex Particle Simulations of a Leapfrogging Vortex Ring, *Computers and Fluids*, Vol. 45, No. 1, pp. 155–161 (2011)
36. Rio Yokota and Shinnosuke Obi, Vortex Methods for the Simulation of Turbulent Flows, *Journal of Fluid Science and Technology*, Vol. 6, No. 1, pp.14–29 (2011)
37. Rio Yokota and Tsuyoshi Hamada, N-body Simulation and FMM on the Large-scale GPU Cluster at Nagasaki University, *Journal of the Japan Society for Computational Engineering and Science*, Vol. 15, No. 4 (2010)
38. Rio Yokota and Shinnosuke Obi, Comparing Vortex Methods and Finite Difference Methods in a Homogeneous Turbulent Shear Flow, *International Journal for Numerical Methods in Fluids*, Vol. 63, No. 7, pp. 828–846. (2010)

39. Rio Yokota, Lorena A. Barba, and Matthew G. Knepley, PetRBF—A Parallel $O(N)$ Algorithm for Radial Basis Function Interpolation with Gaussians, *Computer Methods in Applied Mechanics and Engineering*, Vol. 199, No. 25-28, pp. 1793–1804. (2010)
40. Rio Yokota, Tetsu Narumi, Ryuji Sakamaki, Shun Kameoka, Shinnosuke Obi, and Kenji Yasuoka, Fast Multipole Methods on a Cluster of GPUs for the Meshless Simulation of Turbulence, *Computer Physics Communications*, Vol. 180, No. 11, pp. 2066–2078. (2009)
41. Tarun K. Sheel, Rio Yokota, Kenji Yasuoka, and Shinnosuke Obi, The Study of Colliding Vortex Rings Using a Special-Purpose Computer and FMM, *Transactions of the Japan Society for Computational Engineering and Science*, 20080003 (2008).
42. Rio Yokota, Tarun K. Sheel, and Shinnosuke Obi, Calculation of Isotropic Turbulence Using a Pure Lagrangian Vortex Method, *Journal of Computational Physics*, Vol. 226, pp.1589–1606. (2007)

CONFERENCE
PROCEEDINGS
(PEER-REVIEWED)

1. Daisuke Nohara, Taishi Nakamura, Rio Yokota, Monotonic and Non-Monotonic Length-Performance Relationships in Reasoning RL, ICML 2026 Workshop AdaptFM, Jul. 2026.
2. Satoki Ishikawa, Sameer Satish Deshmukh, Sakina Fatima, Takumi Honda, Rio Yokota, Loss and Optimizer as Two Essential Mechanisms Behind Knowledge Distillation, ICML Workshop HiLD, Jul. 2026.
3. Masaki Kawamura, Nakamasa Inoue, Rintaro Yanagi, Hirokatsu Kataoka, Rio Yokota, PowerCLIP: Powerset Alignment for Fine-Grained Contrastive Pre-Training, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2026.
4. Akiko Aizawa, Yuki Arase, Fei Cheng, Jiahao Huang, Zhiyi Huang, Junfeng Jiang, Teruhito Kanazawa, Daisuke Kawahara, Kazuma Kobayashi, Takashi Kodama, Sadao Kurohashi, Yusuke Oda, Yuma Tsuta, Zhen Wan, Zhishen Yang, Rio Yokota, Building Effective Japanese Medical LLMs with an Open Recipe for Domain Adaptation through Continued Pre-training, International Conference on Language Resources and Evaluation (LREC), May 2026.
5. Taishi Nakamura, Satoki Ishikawa, Masaki Kawamura, Takumi Okamoto, Daisuke Nohara, Jun Suzuki, Rio Yokota, Optimal Sparsity of Mixture-of-Experts Language Models for Reasoning Tasks, International Conference on Learning Representations (ICLR), Apr. 2026.
6. Kazuki Fujii, Yukito Tajima, Sakae Mizuki, Masaki Kawamura, Hinari Shimada, Taihei Shiotani, Koshiro Saito, Masanari Oi, Taishi Nakamura, Takumi Okamoto, Shigeki Ishida, Kakeru Hattori, Youmi Ma, Hiroya Takamura, Rio Yokota, Jun Sakuma, Naoaki Okazaki, Rewriting Pre-Training Data Boosts LLM Performance in Math and Code, International Conference on Learning Representations (ICLR), Apr. 2026.
7. Kazuki Fujii, Rio Yokota, Swallow LLM: Continual Pre-Training and RL for Sovereign AI, GPU Technology Conference (GTC), Mar. 2026.
8. Kazuki Fujii, Yukito Tajima, Masaki Kawamura, Sakae Mizuki, Taishi Nakamura, Hinari Shimada, Masanari Ohi, Taihei Shiotani, Koshiro Saito, Takumi Okamoto, Shigeki Ishida, Youmi Ma, Hiroya Takamura, Rio Yokota, Naoaki Okazaki, Qwen3-Swallow: Continual Pre-Training of Japanese–English Thinking LLMs with Megatron-LM, GPU Technology Conference, Mar. 2026.
9. Hiro Ishii, Kenta Niwa, Hiroshi Sawada, Akinori Fujino, Noboru Harada, Rio Yokota, FedPM: Federated Learning Using Preconditioned Mixing of Local Parameters, The 40th Annual AAAI Conference on Artificial Intelligence, Jan. 2026.
10. Akihiro Ida, Kazuya Goto, Rio Yokota, Tasuku Hiraishi, Toshihiro Hanawa, Takeshi Iwashita, Masatoshi Kawai, Satoshi Ohshima, Tetsuya Hoshino, Tensor-Core-Optimized Strategies for $BLR \times Tall$ -Skinny Matrix Multiplication in BEM, SCA/HPCAsia, Jan. 2026.

11. Yukito Tajima, Nakamasa Inoue, Yusuke Sekikawa, Ikuro Sato, Rio Yokota, Masked Gated Linear Unit, Annual Conference on Neural Information Processing Systems (NeurIPS), Dec. 2025.
12. Avrajit Ghosh, Bai Cong, Rio Yokota, Saiprasad Ravishankar, Rongrong Wang, Molei Tao, Mohammad Emtiyaz Khan, Thomas Möllenhoff, Variational Learning Finds Flatter Solutions at the Edge of Stability, Annual Conference on Neural Information Processing Systems (NeurIPS), Dec. 2025.
13. Satoshi Ohshima, Akihiro Ida, Masatoshi Kawai, Takeshi Fukaya, Rio Yokota, A Study on the Performance and Usability of Managed Memory and Unified Memory for Accelerating Numerical Calculation Program, 18th IEEE International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc), Dec. 2025.
14. Kazuma Kobayashi, Zhen Wan, Fei Cheng, Yuma Tsuta, Xin Zhao, Junfeng Jiang, Jiahao Huang, Zhiyi Huang, Yusuke Oda, Rio Yokota, Yuki Arase, Daisuke Kawahara, Akiko Aizawa, Sadao Kurohashi, Leveraging High-Resource English Corpora for Cross-lingual Domain Adaptation in Low-Resource Japanese Medicine via Continued Pre-training, EMNLP Findings, Nov. 2025.
15. Youmi Ma, Sakae Mizuki, Kazuki Fujii, Taishi Nakamura, Masanari Ohi, Hinari Shimada, Taihei Shiotani, Koshiro Saito, Koki Maeda, Kakeru Hattori, Takumi Okamoto, Shigeki Ishida, Rio Yokota, Hiroya Takamura, Naoaki Okazaki, Building Instruction-Tuning Datasets from Human-Written Instructions with Open-Weight Large Language Models, Conference on Language Modeling (COLM), Oct. 2025.
16. Koshiro Saito, Sakae Mizuki, Masanari Ohi, Taishi Nakamura, Taihei Shiotani, Koki Maeda, Youmi Ma, Kakeru Hattori, Kazuki Fujii, Takumi Okamoto, Shigeki Ishida, Hiroya Takamura, Rio Yokota, Naoaki Okazaki, Why We Build Local Large Language Models: An Observational Analysis from 35 Japanese and Multilingual LLMs, COLM 2025 Workshop on Multilingual and Equitable Language Technologies (MELT), Oct. 2025.
17. Satoki Ishikawa, Tal Ben-Nun, Brian Van Essen, Rio Yokota, Nikoli Dryden, Lion Cub: Minimizing Communication Overhead in Distributed Lion, ICML 2025 Workshop TTODLer-FM, July 2025.
18. Taishi Nakamura, Satoki Ishikawa, Masaki Kawamura, Takumi Okamoto, Daisuke Nohara, Jun Suzuki, Rio Yokota, Optimal Sparsity of Mixture-of-Experts Language Models for Reasoning Tasks, ICML 2025 2nd AI for Math Workshop, July 2025.
19. Chen Zhuang, Lingqi Zhang, Du Wu, Peng Chen, Jiajun Huang, Xin Liu, Rio Yokota, Nikoli Dryden, Toshio Endo, Satoshi Matsuoka, Mohamed Wahib, Scaling Large-scale GNN Training to Thousands of Processors on CPU-based Supercomputers, 39th ACM International Conference on Supercomputing (ICS), June 2025.
20. Thomas Spendlhofer, Qianxiang Ma, Yasuhiro Matsumoto, Rio Yokota, On the Interplay Between Precision, Rank, Admissibility, and Iterative Refinement for Hierarchical Low-Rank Matrix Solvers, ISC High Performance, June 2025.
21. Qi Sun, Edoardo Cetin, Yujin Tang, Transformer² Self-adaptive LLMs, The 13th International Conference on Learning Representations (ICLR), Apr. 2025.
22. Edoardo Cetin, Qi Sun, Tianyu Zhao, Yujin Tang, An Evolved Universal Transformer Memory, The 13th International Conference on Learning Representations (ICLR), Apr. 2025.
23. Satoki Ishikawa, Ryo Karakida, Rio Yokota, Local Loss Optimization in the Infinite Width: Stable Parameterization of Predictive Coding Networks and Target Propagation, The 13th International Conference on Learning Representations (ICLR), Apr. 2025.
24. Satoki Ishikawa, Makoto Yamada, Han Bao, Yuki Takezawa. PhiNets: Brain-inspired Non-contrastive Learning Based on Temporal Prediction Hypothesis, The 13th International Conference on Learning Representations (ICLR), Apr. 2025.

25. Taishi Nakamura, Takuya Akiba, Kazuki Fujii, Yusuke Oda, Rio Yokota, Jun Suzuki, Drop Upcycling: Training Sparse Mixture of Experts with Partial Re-initialization, The 13th International Conference on Learning Representations (ICLR), Apr. 2025.
26. So Kuroki, Taishi Nakamura, Takuya Akiba, Yujin Tang. Agent Skill Acquisition for Large Language Models via CycleQD, The 13th International Conference on Learning Representations (ICLR), Apr. 2025.
27. Qi Sun, Marc Pickett, Aakash Kumar Nain, Llion Jones, Transformer Layers as Painters, The 39th Annual AAAI Conference on Artificial Intelligence (AAAI), Feb. 2025.
28. Chen Zhuang, Peng Chen, Xin Liu, Nikoli Dryden, Rio Yokota, Toshio Endo, Satoshi Matsuoka, Mohamed Wahib, SuperGCN: General and Scalable Framework for GCN Training on CPU-powered Supercomputers, PPOPP, poster, Mar. 2025.
29. Shun Iwase, Shuya Takahashi, Nakamasa Inoue, Rio Yokota, Ryo Nakamura, Hirokatsu Kataoka, Eisaku Maeda, On the Relationship Between Double Descent of CNNs and Shape/Texture Bias Under Learning Process, 27th International Conference on Pattern Recognition (ICPR), Dec. 2024.
30. Bai Cong, Nico Daheim, Yuesong Shen, Daniel Cremers, Rio Yokota, Mohammad Emtiyaz Khan, Thomas Möllenhoff, Variational Low-Rank Adaptation Using IVON, NeurIPS 2024 Workshop on Fine-Tuning in Modern Machine Learning (FITML), Dec. 2024.
31. Kazuki Fujii, Taishi Nakamura, Rio Yokota, llm-recipes, A Framework for Seamless Integration and Efficient Continual Pre-Training of Large Language Models, SC'24 TPC Workshop, Nov. 2024.
32. Taishi Nakamura, Mayank Mishra, Simone Tedeschi, Yekun Chai, Jason T Stillerman, Felix Friedrich, Prateek Yadav, Tanmay Laud, Vu Minh Chien, Terry Yue Zhuo, Diganta Misra, Ben Bogin, Xuan-Son Vu, Marzena Karpinska, Arnav Varma Dantuluri, Wojciech Kusa, Tommaso Furlanello, Rio Yokota, Niklas Muennighoff, Suhas Pai, Tosin Adewumi, Veronika Laipala, Xiaozhe Yao, Adalberto Junior, Alpay Ariyak, Aleksandr Drozd, Jordan Clive, Kshitij Gupta, Liangyu Chen, Qi Sun, Ken Tsui, Noah Persaud, Nour Fahmy, Tianlong Chen, Mohit Bansal, Nicolo Monti, Tai Dang, Ziyang Luo, Tien-Tung Bui, Roberto Navigli, Virendra Mehta, Matthew Blumberg, Victor May, Huu Nguyen, Sampo Pyysalo, Aurora-M: The First Open Source Multilingual Language Model Red-teamed according to the U.S. Executive Order, The 31st International Conference on Computational Linguistics (COLING), Industry Track, Jan. 2025.
33. Naoaki Okazaki, Kakeru Hattori, Shota Hirai, Hiroki Iida, Masanari Ohi, Kazuki Fujii, Taishi Nakamura, Mengsay Loem, Rio Yokota, Sakae Mizuki, Building a Large Japanese Web Corpus for Large Language Models, Conference on Language Modeling COLM, Oct. 2024.
34. Kazuki Fujii, Taishi Nakamura, Mengsay Loem, Hiroki Iida, Masanari Ohi, Kakeru Hattori, Hirai Shota, Sakae Mizuki, Rio Yokota, Naoaki Okazaki, Continual Pre-Training for Cross-Lingual LLM Adaptation: Enhancing Japanese Language Capabilities, Conference on Language Modeling COLM, Oct. 2024.
35. Go Otani, Ryu Tadokoro, Ryosuke Yamada, Yuki Asano, Iro Laina, Christian Rupprecht, Nakamasa Inoue, Rio Yokota, Hirokatsu Kataoka, Yoshimitsu Aoki, Rethinking Image Super-Resolution from Training Data Perspectives, European Conference on Computer Vision (ECCV), Sep. 2024.
36. Ryo Nakamura, Ryu Tadokoro, Ryosuke Yamada, Yuki Asano, Iro Laina, Christian Rupprecht, Nakamasa Inoue, Rio Yokota, Hirokatsu Kataoka, Scaling Backwards: Minimal Synthetic Pre-training?, European Conference on Computer Vision (ECCV), Sep. 2024.
37. Ryosuke Yamada, Kensho Hara, Hirokatsu Kataoka, Koshi Makihara, Nakamasa Inoue, Rio Yokota, Yutaka Satoh, Formula-Driven Visual-Geometric Pre-training, European Conference on Computer Vision (ECCV), Sep. 2024.

38. Ryo Hayamizu, Shota Nakamura, Sora Takashima, Hirokatsu Kataoka, Ikuro Sato, Nakamasa Inoue, Rio Yokota, SIFTer: Self-improving Synthetic Datasets for Pre-training Classification Models, CVPR SynData Workshop, Jun. 2024.
39. Yuesong Shen, Nico Daheim, Gian Maria Marconi, Peter Nickl, Bai Cong, Bazan Clement Emile Marcel Raoul, Rio Yokota, Iryna Gurevych, Daniel Cremers, Mohammad Emtiyaz Khan, Thomas Möllenhoff, Variational Learning is Effective for Large Deep Networks, The 41st International Conference on Machine Learning (ICML), Jul. 2024.
40. Hiromichi Kobayashi, Satoshi Yui, Makoto Tsubota, Tomokazu Saito, Rio Yokota, Interaction between quantum turbulence and normal-fluid turbulence in superfluid helium, Thirteenth International Symposium on Turbulence and Shear Flow Phenomena (TSFP13), Jun. 2024.
41. Satoki Ishikawa, Ryo Karakida, On the Parameterization of Second-Order Optimization Effective towards the Infinite Width, The 12th International Conference on Learning Representations (ICLR), May. 2024.
42. Satoki Ishikawa, Rio Yokota, When Does Second-Order Optimization Speed Up Training? The 12th International Conference on Learning Representations (ICLR), Tiny paper, May. 2024.
43. Ryo Nakamura, Sora Takashima, Edgar Josafat Martinez-Noriega, Nakamasa Inoue, Hirokatsu Kataoka, Rio Yokota, Pre-training Vision Transformers with Very Limited Synthesized Images, IEEE/CVF International Conference on Computer Vision (ICCV), Oct. 2023.
44. Risa Shinoda, Ryo Hayamizu, Kodai Nakashima, Nakamasa Inoue, Rio Yokota, Hirokatsu Kataoka, SegRCDB: Semantic Segmentation via Formula-Driven Supervised Learning, IEEE/CVF International Conference on Computer Vision (ICCV), Oct. 2023.
45. Muhammad Ridwan Apriansyah, Rio Yokota, Computing Eigenvalue of Symmetric $\mathcal{H}^2$ -Matrices in Linear Time with Slicing the Spectrum, International Conference on Parallel Processing (ICPP), Aug. 2023.
46. Sameer Deshmukh, Rio Yokota, George Bosilca, $O(N)$ Distributed Direct Factorization of Structured Dense Matrices Using Runtime Systems, International Conference on Parallel Processing (ICPP), Aug. 2023.
47. Hiroyuki Ootomo, Rio Yokota, Mixed-Precision Random Projection for RandNLA on Tensor Cores, Platform for Advanced Scientific Computing (PASC), Jun. 2023.
48. Sora Takashima, Ryoh Hayamizu, Nakamasa Inoue, Hirokatsu Kataoka, Rio Yokota, Visual Atoms: Pre-training Vision Transformers with Sinusoidal Waves, IEEE/CVF Conference on Computer Vision and Pattern Recognition, Jun. 2023.
49. Hiroyuki Ootomo, Hidetaka Manabe, Kenji Harada, Rio Yokota, Quantum Circuit Simulation by SGEMM Emulation on Tensor Cores and Automatic Precision Selection, ISC High Performance, May 2023.
50. Hiroyuki Ootomo, Rio Yokota, Reducing Shared Memory Footprint to Leverage High Throughput on Tensor Cores and its Flexible API Extension Library, HPC Asia, Feb. 2023.
51. Satoshi Ohshima, Akihiro Ida, Rio Yokota and Ichitaro Yamazaki, QR Factorization of Block Low-Rank Matrices on Multi-Instance GPU, The 23rd International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT'22), Dec. 2022.
52. Hiroki Naganuma, Kartik Ahuja, Ioannis Mitliagkas, Shiro Takagi, Tetsuya Motokawa, Rio Yokota, Kohta Ishikawa, Ikuro Sato, Empirical Study on Optimizer Selection for Out-of-Distribution Generalization, NeurIPS Workshop Distshift, Dec. 2022.
53. Kazuki Osawa, Satoki Ishikawa, Rio Yokota, Shigang Li, Torsten Hoefer, ASDL: A Unified Interface for Gradient Preconditioning in PyTorch, NeurIPS Workshop Order up! The Benefits of Higher-Order Optimization in Machine Learning, Dec. 2022.
54. Aoyu Li, Ikuro Sato, Kohta Ishikawa, Rei Kawakami, Rio Yokota, Informative Sample-Aware Proxy for Deep Metric Learning, ACM Multimedia Asia, Dec. 2022. (Best paper)

55. Qianxiang Ma, Sameer Deshmukh, Rio Yokota, Scalable Linear Time Dense Direct Solver for 3-D Problems Without Trailing Sub-Matrix Dependencies, The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC22), Nov. 2022.
56. Hirokatsu Kataoka, Ryo Hayamizu, Ryosuke Yamada, Kodai Nakashima, Sora Takashima, Xinyu Zhang, Edgar Josafat Martinez-Noriega, Nakamasa Inoue, Rio Yokota, Replacing Labeled Real-image Datasets with Auto-generated Contours, IEEE/CVF Conference on Computer Vision and Pattern Recognition, Jun. 2022.
57. Hana Hoshino, Kei Ota, Asako Kanezaki, Rio Yokota, OPIRL: Sample Efficient Off-Policy Inverse Reinforcement Learning via Distribution Matching, IEEE International Conference on Robotics and Automation, May 2022.
58. Shun Iwase, Xingyu Liu, Rawal Khirodkar, Rio Yokota, Kris M. Kitani, RePOSE: Real-Time Iterative Rendering and Refinement for 6D Object Pose Estimation, International Conference on Computer Vision, Oct. 2021.
59. Hikaru Nakata, Nakamasa Inoue, Rio Yokota, Self-supervised Continual Pretraining for Class Incremental Image Classification, Proc. CVPR CLVISION Workshop (Findings), Jun. 2021.
60. Ryo Karakida, Kazuki Osawa, Understanding Approximate Fisher Information for Fast Convergence of Natural Gradient Descent in Wide Neural Networks, Advances in Neural Information Processing Systems (NeurIPS), oral presentation, Dec. 2020.
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JAPANESE
DOMESTIC
CONFERENCE (NOT
PEER-REVIEWED)

1. 安田俊吾, 横田理央, 視覚・言語モデルにおける日本語 OCR 性能の向上, 情報処理学会全国大会, Mar. 2026.
2. 野原大輔, 中村泰士, 横田理央, 大規模言語モデルの強化学習による効率的な論理推論, 情報処理学会全国大会, Mar. 2026.
3. 岡本拓己, 横田理央, One-Shot NAS による BERT のモデル圧縮, 人工知能学会全国大会 (第 38 回), May 2024.
4. 中村泰士, 横田理央, 継続学習を用いた効率の良いマルチリンガル・マルチエキスパートモデルの開発, 情報処理学会全国大会, Mar. 2024.
5. 藤井一喜, 横田理央, 大規模言語モデルの分散並列学習, 情報処理学会全国大会, Mar. 2024.
6. 岡本拓己, 横田理央, 大規模言語モデルの構造探索, 情報処理学会全国大会, Mar. 2024.
7. 岡崎直観, 服部翔, 平井翔太, 飯田大貴, 大井聖也, 藤井一喜, 中村泰士, Mengsay Loem, 横田理央, 水木栄, Swallow コーパス: 日本語大規模ウェブコーパス, 言語処理学会第 30 回年次大会, Mar. 2024.
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14. 齋藤智和, 横田理央, 量子渦計算の高速多重極展開法を用いた高速化, 情報処理学会全国大会, Mar 2023.

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25. 石井央, 横田理央, 深層学習における 2 次最適化の汎化性能の検証, 第 84 回情報処理学会全国大会, Mar. 2022.
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34. 所畑貴大, 長沼大樹, 横田理央, 確率的重み付け平均法のラージバッチ学習における有用性の検証, 情報処理学会全国大会, Mar. 2020.
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36. 横田理央, 二次最適化を用いた ImageNet の大規模分散深層学習, 精密工学会 画像応用技術専門委員会, 第 5 回定例研究会「人工知能・データサイエンス」, Jan. 2020.

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39. 横田理央, 深層学習に現れる密行列も構造, 自動チューニング研究会マイクロワークショップ, Oct. 2019.
40. 横田理央, 深層学習の高速化と大規模並列化, 情報処理学会連続セミナー第 3 回: AI と歩む未来 (2): 画像・映像処理の最前線, Sep. 2019
41. Peter Spalthoff, 横田理央. Flexible and Simplistic Hierarchical Matrix-Based Fast Direct Solver, 第 170 回ハイパフォーマンスコンピューティング研究発表, Jul. 2019.
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43. 横田理央, ImageNet ベンチマークの大規模並列深層学習, 第 14 回名工大・核融合研共同セミナー: 流体シミュレーションとディープラーニング, Jul. 2019.
44. 横田理央, 階層的低ランク近似法の最新研究動向と応用例, 第 22 回 AT 研究会オープンアカデミックセッション, May. 2019.
45. 横田理央, 大沢和樹, 辻陽平, 上野裕一郎, 成瀬彰. 大規模並列深層学習における 2 次の最適化手法の効果, 電子情報通信学会総合大会, Mar. 2019.
46. 長沼大樹, 横田理央. 大規模並列深層学習のための目的関数の平滑化, 第 81 回情報処理学会全国大会, Mar. 2019.
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53. 大友広幸, 大沢和樹, 横田理央. フィッシャー情報行列のクロネッカー因子分解を用いた深層学習, 情報処理学会全国大会, Mar. 2018.
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61. 横田理央, 小尾晋之介. 渦法による一様等方性乱流の解析, 第 20 回数値流体力学シンポジウム, Dec. 2006.

CONFERENCE
PROGRAM
COMMITTEE
MEMBER

1. International Conference on Parallel Architectures and Compilation Techniques (PACT 2025), publicity chair
2. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2025), proceedings vice-chair, program committee, AI4S committee
3. Conference on Neural Information Processing Systems (NeurIPS 2025), reviewer
4. IEEE International Conference on Cluster Computing (IEEE CLUSTER 2025), track co-chair
5. International Conference on Machine Learning (ICML 2025), reviewer
6. ISC High Performance (ISC 2025), program committee
7. Platform for Advanced Scientific Computing (PASC 2025), domain co-chair
8. 39th IEEE International Parallel and Distributed Processing Symposium, (IPDPS 2025), program committee, best OSS judge
9. The IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR 2024), reviewer
10. 38th Conference on Neural Information Processing Systems (NeurIPS 2024), reviewer
11. The 12th International Conference on Learning Representations (ICLR 2024), reviewer
12. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2024), poster, ML track, AI4S/TPC WS, LLMs for HPC BoF
13. International Conference on Preconditioning Techniques for Scientific and Industrial Applications, (Precond 2024), program committee
14. 38th IEEE International Parallel and Distributed Processing Symposium, (IPDPS 2024), track co-chair
15. ISC High Performance (ISC 2024), posters chair
16. The 30th International Conference on Parallel Processing (Euro-Par 2024), program committee
17. ACM International Symposium on High-Performance, Parallel and Distributed Computing (HPDC 2024), program committee
18. SC Asia (SCA 2024), program committee
19. Platform for Advanced Scientific Computing (PASC 2024), program committee
20. SIAM Conference on Linear Algebra (LA24), scientific committee
21. SIAM Conference on Computational Science and Engineering (CSE23), poster judge
22. 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023), program committee
23. IEEE International Conference on Cluster Computing (IEEE CLUSTER 2023), publicity chair
24. Principles and Practice of Parallel Programming (PPoPP 2023), publicity chair
25. 37th IEEE International Parallel and Distributed Processing Symposium, (IPDPS 2023), program committee

26. ISC High Performance (ISC 2023), posters deputy chair
27. Platform for Advanced Scientific Computing (PASC 2023), mini symposia and posters committee, program committee
28. The 23rd IEEE/ACM international Symposium on Cluster, Cloud and Internet Computing (CCGrid 2023), program committee
29. The IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR 2023), reviewer
30. The 32nd International Conference on Parallel Architectures and Compilation Techniques (PACT 2023), publicity chair Asia
31. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2023), posters, workshops, ML tech paper, panelist, best paper
32. 37th Conference on Neural Information Processing Systems (NeurIPS 2023), reviewer
33. The Eleventh International Conference on Learning Representations (ICLR 2023), reviewer
34. European Conference on Computer Vision (ECCV 2022), reviewer
35. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2022), program committee, workshop chair
36. 49th International Conference on Parallel Processing (ICPP 2022), program committee
37. IEEE Cluster (IEEE Cluster 2022), program committee
38. ISC High Performance (ISC 2022), poster committee
39. The 36th IEEE International Parallel & Distributed Processing Symposium (IPDPS 2022), program committee
40. International Conference on High Performance Computing in Asia-Pacific Region (HPC Asia 2022), Applications track co-chair
41. IEEE International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), program committee
42. 48th International Conference on Parallel Processing (ICPP 2021), program committee
43. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2021), program committee
44. IEEE Cluster (IEEE Cluster 2021), program committee
45. ISC High Performance (ISC 2021), program committee, best paper committee, steering committee
46. The 35th IEEE International Parallel & Distributed Processing Symposium (IPDPS 2021), program committee
47. International Conference on High Performance Computing in Asia-Pacific Region (HPC Asia 2021), organizing committee
48. SIAM Computational Science and Engineering (SIAM CSE 2021), organizing committee
49. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2020), ML track Chair
50. The 26th International Conference on Parallel Processing (Euro-Par 2020), program committee
51. SIAM Conference on Parallel Processing for Scientific Computing (SIAM PP 2020), program committee
52. International Conference on High Performance Computing in Asia-Pacific Region (HPC Asia 2020), program committee
53. The 3rd cross-disciplinary Workshop on Computing Systems, Infrastructures, and Programming (xSIG 2019), program committee

54. IEEE 13th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc 2019), ATMG Track Chair
55. 48th International Conference on Parallel Processing (ICPP 2019), Applications track chair, Workshop co-organizer
56. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2019), program committee
57. ISC High Performance (ISC 2019), program committee
58. IEEE Cluster (IEEE Cluster 2019), program committee
59. Platform for Advanced Scientific Computing Conference (PASC 2019), program committee
60. SIAM Conference on Computational Science and Engineering (SIAM CSE19), organizing committee
61. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2018), proceedings chair
62. ISC High Performance (ISC 2018), proceedings chair
63. IEEE Cluster (IEEE Cluster 2018), program committee
64. The 32nd IEEE International Parallel & Distributed Processing Symposium (IPDPS 2018), program committee
65. 18th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid 2018), program committee
66. SC Asia (SCA 2018), program committee
67. SIAM Conference on Parallel Processing for Scientific Computing (SIAM PP 2018), organizing committee
68. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2017), program committee
69. Platform for Advanced Scientific Computing Conference (PASC 2017), program committee
70. The 31st IEEE International Parallel & Distributed Processing Symposium (IPDPS 2017), program committee
71. International Conference on Supercomputing (ICS 2017), program committee
72. IEEE Cluster (IEEE Cluster 2017), program committee
73. The 23rd International Conference on Parallel Processing (Euro-Par 2017), program committee
74. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2016), program committee
75. Platform for Advanced Scientific Computing Conference (PASC 2016), program committee
76. ISC High Performance (ISC 2016), program committee
77. The 30th IEEE International Parallel & Distributed Processing Symposium (IPDPS 2016), program committee
78. The 23rd annual IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC 2016), program committee
79. 16th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid 2016), program committee
80. International Conference on Supercomputing (ICS 2016), program committee
81. IEEE Cluster (IEEE Cluster 2016), program committee
82. 28th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD 2016), program committee

83. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2015), program committee
84. The 22nd annual IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC 2015), program committee
85. 15th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid 2015), program committee
86. IEEE Cluster (IEEE Cluster 2015), program committee
87. The 20th International Conference on Parallel Processing (Euro-Par 2014), program committee
88. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2014), program committee
89. The International Meeting on High-Performance Computing for Computational Science (VECPAR 2014), program committee
90. The annual IEEE International Conference on High Performance Computing (HiPC 2014), program committee
91. 14th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid 2014), program committee
92. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC 2013), program committee
93. The 28th IEEE International Parallel & Distributed Processing Symposium (IPDPS 2013), program committee
94. 18th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP 2013), program committee
95. The annual IEEE International Conference on High Performance Computing (HiPC 2013), program committee
96. 15th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC 2013), program committee
97. The 12th International Symposium on Parallel and Distributed Computing (ISPDC 2012), program committee
98. The Third International Workshop on Frontier of GPU Computing (FCG 2012), program committee

RESEARCH GRANTS

1. 2025–2028. Renovation of science and technology software to unlock the potential of next-generation computers (Grant-in-Aid for Scientific Research (A), JSPS).
2. 2024–2027. Expanding the applicability of low-rank structured matrix methods and leveraging diverse computing architectures (Grant-in-Aid for Scientific Research (B), JSPS).
3. 2024–2027. Research toward establishing an outcome-extraction methodology for evaluating the efficacy and safety of pharmaceuticals using Large Language Models (LLM) (Health, Labour and Welfare Sciences Research Grant, Applied Research, MHLW).
4. 2023–2027. Next-generation high-performance linear solver for the future of computational science and engineering (Grant-in-Aid for Scientific Research (A), JSPS).
5. 2023–2026. Compositional pattern recognition and understanding using deep generative models (Grant-in-Aid for Scientific Research (A), JSPS).
6. 2022–2026. Functional improvement of mixing promotion in fluid equipment by elucidating the universal statistical law of two-phase turbulence (Grant-in-Aid for Scientific Research (B), JSPS).

7. 2022–2025. Fast and accurate eigenvalue calculations by hierarchical low-rank approximation and its application to large-scale electronic structure calculations (Grant-in-Aid for Scientific Research (B), JSPS).
8. 2021–2027. A New Bayes Duality Principle for Adaptive, Robust, Life-long Learning of AI (CREST, JST).
9. 2021–2024. Construction of numerical linear algebra based on lattice H-matrices and its high-performance implementation on modern architectures (Grant-in-Aid for Scientific Research (B), JSPS).
10. 2020–2023. Application of unconventional linear algebra techniques to continual learning in supergiant neural networks (Grant-in-Aid for Challenging Research (Pioneering), JSPS).
11. 2020–2023. Life-Long Deep Learning using Bayesian Principles (Grant-in-Aid for Scientific Research (B), JSPS).
12. 2018–2021. Linear solvers for machine learning hardware (Grant-in-Aid for Scientific Research (B), JSPS).
13. 2017–2020. Enhancement of an H-matrix library and optimization for next-generation supercomputers (Grant-in-Aid for Scientific Research (B), JSPS).
14. 2016–2022. Acceleration and economization of deep learning algorithms for image processing in social infrastructure (CREST, JST).
15. 2016–2019. Hierarchical particle-method framework for exascale computers balancing performance and productivity (Grant-in-Aid for Scientific Research (B), JSPS).
16. 2016–2018. Large-scale iterative solvers combining the FMM and H-matrices (Grant-in-Aid for Young Scientists (A), JSPS).
17. 2015–2016. Algebraic extension of the FMM as a preconditioner for exascale large-scale linear systems (Grant-in-Aid for Research Activity Start-up, JSPS).